

Equirectangular ChArUco Calibration Report

Step-by-step summary of the current calibration pipeline.

Goal: calibrate an omnidirectional / fisheye camera using a ChArUco board.

Pipeline:

1. Sample raw frames from the calibration video.
2. Detect ChArUco markers and corners in raw frames.
3. Use high-quality raw detections for preliminary calibration.
4. Rectify useful raw frames with the preliminary calibration.
5. Detect ChArUco corners again on rectified frames.
6. Use the cleaner rectified detections for final calibration.

Final RMS reprojection error: 0.869140

Final calibrated frames: 35

Step 1 - Raw frame detection

Input video frames were sampled every 30 frames.

Sampled frames: 372

Frames with markers: 274

Frames marked useful: 67

Average ChArUco corners in useful raw frames: 6.51

Outputs:

data/raw_frames/

data/detected_frames/

results/detection_summary.csv

Step 2 - Preliminary calibration

The preliminary calibration uses stricter raw detections, requiring at least 9 ChArUco corners per frame.

Frames used: 19

RMS reprojection error: 1.893938

Camera matrix:

```
[[7.865689e+02 0.000000e+00 1.002871e+03]
```

```
[0.000000e+00 7.893097e+02 5.274686e+02]
```

```
[0.000000e+00 0.000000e+00 1.000000e+00]]
```

Distortion coefficients:

```
[-0.274188 0.078393 0.001685 -0.002818 -0.01009 ]
```

Output:

results/preliminary_calibration.npz

Step 3 - Rectification and second-pass detection

Useful raw frames are rectified using the preliminary calibration. Full-frame rectification is used with $\alpha=1$ to preserve more board area.

Rectified frames checked: 67

Rectified frames with markers: 67

Rectified frames useful for final calibration: 35

Average ChArUco corners in useful rectified frames: 7.69

Outputs:

data/rectified_frames/

data/rectified_detected_frames/

results/rectification_summary.csv

results/rectified_detection_summary.csv

Step 4 - Final calibration

The final calibration uses rectified detections. This pass uses at least 6 ChArUco corners per frame because detections are cleaner after rectification.

Frames used: 35

RMS reprojection error: 0.869140

RMS improvement from preliminary to final: 54.11%

Camera matrix:

```
[[4.475080e+02 0.000000e+00 1.225324e+03]
```

```
[0.000000e+00 2.787269e+02 5.181065e+02]
```

```
[0.000000e+00 0.000000e+00 1.000000e+00]]
```

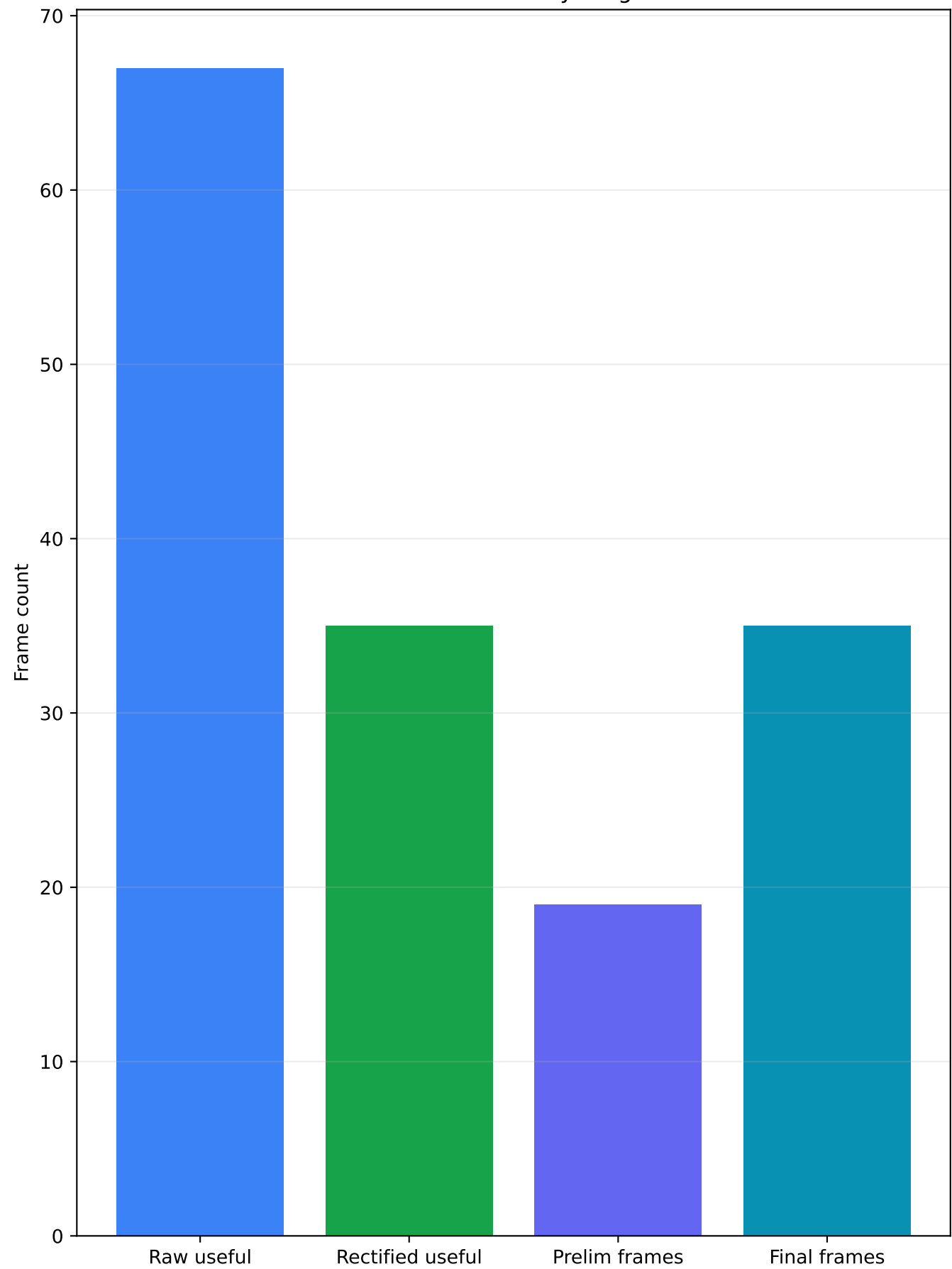
Distortion coefficients:

```
[-0.031815 0.027696 -0.000504 -0.006176 -0.008445]
```

Output:

results/final_calibration.npz

Frame Counts by Stage



Example Raw ChArUco Detection



Example Rectified ChArUco Detection

